

SURVEY

Machine Learning Based Solar Photovoltaic Power Forecasting: A Review and Comparison

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ABSTRACT The growing interest in renewable energy and the falling prices of solar panels place solar electricity in a favourable position for adoption. However, the high-rate adoption of intermittent renewable energy introduces challenges and the potential to create power instability between the available power generation and the load demand. Hence, accurate solar Photovoltaic (PV) power forecasting is essential to maintain system reliability and maximize renewable energy integration. The current solar PV power forecasting approaches are an essential tool to maintain system reliability and maximize renewable energy integration. This paper presents a comprehensive and comparative review of existing Machine Learning (ML) based approaches used in PV power forecasting, focusing on short-term horizons. We provide an overview of factors affecting solar PV power forecasting and an overview of existing PV power forecasting methods in the literature, with a specific focus on ML-based models. To further enhance the comparison and provide more insights into the advancement in the area, we simulate the performance of different ML methods used in solar PV power forecasting and, finally, a discussion on the results of the work.

INDEX TERMS Artificial intelligence, artificial neural networks, photovoltaic, power forecasting, renewable energy.

I. INTRODUCTION

Electricity has become a significant part of the modern world. Its demand is predicted to rise as the world becomes more globalized and modernized [1]. Electricity generation from fossil fuel-based power plants has resulted in significant pollution in terms of Carbon dioxide (CO₂) emissions and greenhouse gas (GHG) emissions, resulting in significant climate change around the planet. Hence alternative renewable energy sources (RESs) such as solar, wind, and biomass

have the potential to play a key role in the future. According to the Global Energy Review in the year 2020, the share of renewable electricity supply contributed close to 28% of global electricity generation in the first quarter of 2020. An increase from 26% in the first quarter of 2019 [2]. The growth is primarily due to the contributions of solar and wind farms, spurred on by the decrease in wind turbines and solar panels cost [3] and also many policies adopted by countries to reduce and regulate CO₂ emissions and global warming [4]. Encouraged by low, declining prices and large contributions to carbon emission reduction, the power sector is progressing increasingly into integrating and using renewable

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energy sources. The use of RE has brought about momentous changes in the field of energy production and has demonstrated the potential of clean and unlimited energy for the future.

There exist several RESs, such as hydropower, bioenergy, wind energy, solar thermal, solar photovoltaic (PV) energy, and many more. Focusing on solar energy, solar photovoltaic (PV) systems have experienced a large growth in electricity generation, with an increased number of installed systems [5], [6], [7], [8]. Such PV systems have been installed in open spaces or rooftops, integrated into buildings and vehicles, and placed in large arrays in solar power plants. However, besides solar PV contributing to diversifying power generation and solving global warming solutions, the high-rate adoption of solar PV introduces challenges.

Solar energy is an intermittent Renewable energy source (RES). Hence, it varies based on both weather conditions and site-specific conditions. The possibility of creating power instability increases in tandem with the increase in the number of renewables in the energy mix [9], [10], increasing the importance of controlling the flow of renewable power.

Solar PV forecasting is the process of predicting the amount of electrical power that will be generated by a solar photovoltaic (PV) system over a specified time. This involves the use of mathematical models to predict the output of a PV system based on a range of input parameters. The ambiguity of the load consumption as well as the intermittent power production of the solar PV production can result in differences between the available power generation and the demanded load. Hence, accurate prediction or forecasting of solar PV power generation capacity is essential.

Accurate solar PV forecasting is important for several reasons. First, it allows grid operators or independent systems to anticipate changes in the supply of electricity and to adjust the distribution of power accordingly. This helps to maintain the stability and reliability of the power grid or independent systems isolated from the grid. Second, accurate forecasts can help to optimize the use of renewable energy sources, reduce the need for backup power sources, and ultimately lower the cost of electricity for consumers.

Solar PV forecasting is a challenging problem due to the complex and dynamic nature of solar energy production. Factors such as cloud cover, humidity, and temperature can all affect the output of a PV system.

As a result, accurate solar PV forecasting requires the use of sophisticated models and the collection of high-quality data. If a forecast is inaccurate, it can result in an imbalance between the available power generation and the demand, which can lead to power outages or other disruptions. In addition, an inaccurate forecast can lead to underutilization of the solar PV systems, which can limit the amount of renewable energy that is integrated into the power grid and increase the reliance on fossil fuel-based sources of electricity.

To address these challenges, researchers and industry professionals have developed a range of solar PV forecasting techniques, which can be broadly classified as physical

(based on physical principles), Data driven (statistical and Machine Learning) or hybrid (Combination of one or more). These techniques use various input variables, such as weather data, historical power generation data, and solar irradiance measurements, to predict the output of solar PV systems.

1) PHYSICAL MODELS (WHITE BOX)

Physical models are based on the fundamental physical principles that govern the behavior of solar PV systems. These models take into account the basic physical properties of the solar PV system, such as the panel efficiency, the angle of incidence of the sun's rays, and the shading of the panels. The models also consider the environmental factors that affect solar energy production, such as temperature, cloud cover, and humidity. Physical models are typically based on mathematical equations that describe the physical behavior of the solar PV system. These equations can be quite complex and require detailed knowledge of the underlying physics and system characteristics (installation details, electrical configuration, geographic location and technical characteristics) of the PV system being modelled [11]. Furthermore, since they utilize weather conditions as input for forecasting, this results in models performance heavily relying on the accuracy of weather forecast data [12].

2) DATA DRIVEN MODELS

Data driven forecasting models refer to models that rely on the use and analysis of historical data to derive insights, patterns, and relationships that can be used to create models to predict the output of a solar PV system. These models are based on statistics and algorithms that can learn patterns and relationships from data and make accurate predictions. However, they require large amounts of high-quality data to be effective, and the accuracy of the predictions can be affected by biases in the data or incomplete data. Data driven can be further subdivided into two categories, these being statistical and machine learning models. The boundaries between statistical modelling and machine learning are often blurry, and some researchers consider them to be the same depending on the specific context and perspective of the user.

a: STATISTICAL MODELS

Statistical models are based on statistical analysis of historical data to identify trends and patterns that can be used to forecast future solar PV power output. These models use statistical techniques and mathematical algorithms to fit historical data and derive relationships between different variables, such as solar irradiance, weather conditions, and energy production. There are many different types of statistical models used in solar PV forecasting, including autoregressive models (AR), moving average models (MA), and Autoregressive Integrated Moving Average (ARIMA). ARIMA models are a class of models that are commonly used for time series analysis and forecasting. They assume that the current value of a time series depends on its past values and

the errors in previous predictions. The ARIMA model uses these relationships to generate predictions for future values of the time series.

While statistical models such as ARIMA models do involve the use of algorithms to identify patterns and relationships, which is a key aspect of machine learning, they are in a separate category from machine learning techniques since the models typically rely on statistical methods such as maximum likelihood estimation, rather than the iterative learning algorithms used in many machine learning techniques.

b: MACHINE LEARNING MODELS (BLACK BOX)

Machine learning models, on the other hand, are based on a more complex set of algorithms that allow the model to learn patterns and relationships in the data on its own. These models can handle more complex and nonlinear relationships between the input variables and output variables and can often generate more accurate forecasts than statistical models. As such, the models have been reported to produce accurate PV forecasting [13]. Examples of machine learning models used in solar PV forecasting include artificial neural networks (ANN), support vector machines (SVM), and random forests (RF).

One key difference between statistical models and machine learning models is the level of human input required. Statistical models generally require more human input in terms of selecting and engineering features, designing the model architecture, and tuning hyperparameters. In contrast, machine learning models can be trained on raw data with minimal preprocessing, and the model architecture and hyperparameters can often be automatically optimized using techniques such as grid search or Bayesian optimization.

Another difference between the two approaches is their ability to handle uncertainty. Statistical models are often more transparent and can provide estimates of uncertainty in their forecasts, whereas machine learning models are often more black-box and may not provide as much information about how the model arrived at its predictions. However, techniques such as Monte Carlo simulation can be used to generate probabilistic forecasts from machine learning models.

3) HYBRID MODELS

Hybrid models are models that combine one or more forecasting methods from either physical or data driven methods with the goal of overcoming the limitations of individual techniques and incorporating the advantages of each method. This does come at a cost in the form of increased computational time.

By using a combination of these techniques and continuously improving the accuracy of solar PV forecasting, it is possible to maintain system reliability, maximize renewable energy integration, and ultimately transition to a more sustainable and carbon-free energy system.

In this context, the scope and objective of this paper is to address accurate solar PV power forecasting as an essential

tool to maintain system reliability and maximize renewable energy integration. This is intended to be achieved by presenting a comprehensive and comparative review on existing Machine Learning (ML) based approaches used in PV power generation. Though there are related literature surveys regarding solar PV power generation forecasting, most do not perform simulation comparisons on short-term forecasts. Therefore, the presented work is different in that emphasis is made on the comparison of short-term horizons and carries out a comparative review with simulations.

The focus contributions of the paper are as follows:

1. We gave an overview of factors affecting solar PV power forecasting regarding temporal classifications and weather.
2. We gave a detailed of Artificial Intelligence (AI) and ML and their application in ML based Solar PV power forecasting.
3. We provided an overview of the performance metrics used for solar PV power forecasting.
4. We provide a comparison of existing solar PV power forecasting methods in the literature, with a focus on ML-based models and a short-term horizon.
5. Finally, we provided an experiment setup and simulation of the performance of different ML methods used in solar PV power forecasting.

The remainder of the paper is arranged as follows: Section II presents an overview of factors affecting solar PV power forecasting regarding temporal classifications and weather. Section III presents an overview of Solar PV power forecasting models with emphasis on ML based models. In Section IV, we provide an overview of the performance metrics used for solar PV power forecasting. Section V gives a comparison of existing PV power forecasting methods in the literature, with a focus on ML-based models and a short-term horizon. In Section VI, experiment setup is provided, while Section VII provides results and discussion on the results. Finally, Section VIII provides the conclusion and future work.

II. FACTORS AFFECTING SOLAR PV POWER FORECASTING

Solar PV forecasting methods face several factors that affect their forecasting performance and accuracy. These factors include forecasting horizon and weather conditions [14], [15].

1) FORECASTING HORIZONS

Several methods are used in literature to perform forecasts of solar PV power generation. The methods can be classified based on temporal aspects, spacial aspects or the data used for the forecast. It is crucial to introduce three ideas with regard to the temporal component of forecasts, these are forecast interval (fi), forecast resolution (fr) and forecast horizon (fh). The temporal span between the current time (t) and the time of the next prediction is known as the forecast horizon. [5]. The frequency at which forecasts are given is described by

TABLE 1. Solar PV power forecasting time horizons.

Time Horizon Classification	Forecasting Time Span
Very Short term	Less than a Day
Short term	A Day to 3 Days
Medium term	Days to Months
Long term	Months to Years

the forecast resolution, while the time span of projections is indicated by the forecast interval. The forecasting procedures of PV power generation are generally categorized in the literature based on their forecast horizons [16], [17], [18], [19]. These are some of the typical forecasting time horizons [16], [17] shown in TABLE 1.

Different applications fall under different forecasting horizons. For example Short term forecast is used in day-ahead energy markets and energy management applications such as energy storage and load scheduling and energy market participation [20]. Not considering the implications of time horizon on forecasting accuracy can result in poor forecasting performance and hence impact the activities that rely on the information. In [21], a 15-90 min time horizon PVPF model was developed. By varying the horizon's length and maintaining all other parameters at constant; the accuracy of the model was found to vary relative to the horizon's length. On the same dataset, similar finding were observed in [22] where a forecast error was found to increase from 3.2% to 15.5% for horizons ranging from 20 to 180 seconds. Other studies have adopted multi-scale hybrid forecast models with different forecast horizons. However, the nRMSE values kept on increasing with horizon's duration regardless of the type of a model employed. This goes to show that forecasting horizons does have a negative impact of forecasting accuracy.

2) WEATHER CONDITIONS

In solar energy harvesting, Photovoltaic (PV) transducers/cells are the most common energy transducers which convert the solar irradiance received on the earth's surface into electrical energy [23], [24]. A solar panel is a transducer made from a combination of parallel and series electrical connections of PV modules, which in turn are made up of connections of PV cells [24]. The efficiency of the cells and hence that of the panels used for solar energy harvesting depends on the amount of solar energy or solar irradiance falling on the panel and the ambient temperature at which the panel is operating [25]. Several other factors, such as the geographical location and orientation of the installation, the materials and design of the panels and the electrical components used such as charge controllers and inverters also influence the amount of energy produced by a PV system. However, once the PV system is installed and operational, these factors remain constant, except for atmospheric conditions which can vary. The output of a PV system is measured under standard test conditions (STC) with a cell temperature of 25 °C, solar intensity of 1000 W/m² and an air mass of 1.5.

However, since atmospheric conditions often differ from the STC, the actual energy output of the PV system may differ significantly from the rated value. Hence weather conditions are a significant determinant of the PV panels' variance and stability [26].

Variations in sun irradiation that reach the earth are what generate the uncertainty and variability in the solar resource, hence impacting PV output power. These variations can be split into deterministic and stochastic components. The motion of the Earth as it rotates around its axis with reference to the sun (the diurnal cycle) and the earth orbiting around the Sun (Annual cycle) provides an explanation for the deterministic aspect. Physical equations are capable of adequately describing such predictable features of sun irradiation [5].

The stochastic behavior of the atmosphere causes greater and unexpected fluctuations in PV output power. These refer to the meteorological or weather conditions. Numerous climatic factors, including cloud cover, humidity level, wind speed, temperature and air pressure contribute to this unpredictability. As such weather forecasting and weather forecasted data can be used to improve solar PV systems power forecasting. Furthermore, depending on where the PV plant is located, different meteorological variables have different correlation factors with PV power. More information on the analysis of weather on PV power forecasting can be found in [27].

III. SOLAR PV POWER FORECASTING MODELS

Utilities that highly depend on PV reliability need to predict both the short and long-term production capabilities of solar PV units. As such, accurate PV energy harvest forecasting is necessary. Generally, PV energy forecasting can be carried out either using the indirect or direct approach. The direct approach performs forecasts that produce PV power as an immediate and final output [28], [29]. Whereas the indirect approach first predicts solar irradiation from meteorological or weather forecast data as input to PV system models to produce the predicted PV power harvest [30]. The indirect approach is preferred where historical PV power data is not available, such as in newly constructed PV systems [31] or where weather forecast data is to be used in other applications.

In literature, there are many techniques that have been studied and used to perform solar PV power forecasting. Below are some of the techniques used for existing solar PV power forecasting.

A. ARTIFICIAL INTELLIGENCE (AI) AND MACHINE LEARNING (ML) IN SOLAR PV POWER FORECASTING

Artificial intelligence (AI) and machine learning (ML) promise to improve modeling and forecasting for a host of problems. In solar PV energy harvests, researchers are adopting AI and ML techniques much more widely since they demonstrate their power in various applications, including bias correction, handling large datasets, reducing cognitive overload, and discovering new knowledge in large data sets.

AI is a field of computer science that focuses on creating computer systems that can demonstrate intelligent behaviors. This is done by programming them to be able to simulate and imitate the thought processes of humans. AI techniques have a range of advantages, such as the ability to quickly generalize data, work with multiple variables, incorporate physical understanding into models, and allow for the discovery of patterns and knowledge from data.

Machine learning (ML) is a subfield under AI involved in the study and building of systems that learn and improve through experience gained from data. In ML, algorithms build models based on sample data, known as training data, to make predictions or decisions without relying on explicitly programmed instructions, formulas, or equations. ML models are presented with many examples relevant to a task. They find statistical structure in these examples that eventually allow the system to develop rules for automating the task. ML-based models are promising solutions for predictions in Renewable energy generation. Furthermore, ML-based models can conduct highly nonlinear problems [32]; as such, they are ideal for fluctuating environments as they can adapt to new data [33]. This gives ML models a significant advantage relative to statistical models.

1) CLASSIFICATION OF ML ALGORITHMS

The learning algorithms or approaches in Machine Learning are typically categorized into supervised [34] and unsupervised [35]. In some cases, a third approach, semi-supervised learning [36] may be considered. Fig. 1 shows a summarized overview and classification of ML algorithms.

a: SUPERVISED LEARNING

Also known as supervised machine learning, it is an approach to machine learning that uses labelled data to train (generate models) and predict outcomes using the knowledge learned from the data. The trained models can be used to assign labels (assign data to specific categories) or make numerical predictions on new data from the problem that is being solved. The bulk of supervised learning involves problems of classification and regression.

b: UNSUPERVISED LEARNING

Contrary to supervised machine learning, unsupervised learning is an approach that uses data that is not labelled to train; therefore, must discover hidden patterns or data groupings without the aid of labels. The algorithms used in unsupervised learning generate models that can analyze, cluster, and categorize unlabeled data as well as detect outliers. Unsupervised learning is used for clustering, association, and dimensional-reduction.

c: SEMI-SUPERVISED LEARNING

Semi-supervised learning, as the name suggests, mixes supervised and unsupervised learning. It is a learning approach that combines the use of both labeled and unlabeled data

to explore problems and generated labeled data for training where most of the training data is not labeled and is paired with a few data that are labeled.

Fig. 2 provides a depiction of the learning or training process in terms of input data, output data and mapping function involved in the ML approaches.

2) ML ALGORITHMS USED IN SOLAR PV POWER FORECASTING MODEL DEVELOPMENT

Since solar PV power forecasting is a regression problem, most of the ML approaches used fall under the category of supervised learning. Below are some of the ML techniques used for solar PV power forecasting:

a: K-NEAREST NEIGHBOURS (K-NN)

K-nearest neighbors (K-NN) is an ML algorithm that can be used for both regression and classification problems. It was first developed by Evelyn Fix and J.L. Hodges in 1951 [37] and expanded upon by [38]. It is based on an algorithm that compares the current state of an unknown data point with the known states of other data points that are like it. These known states are called “training samples” and are represented in a feature space, which is a mathematical representation of the data. To make a prediction using the K-NN algorithm, we first calculate the Euclidean distances between the unknown data point and the training samples in the feature space. Then, we select the first “k” training samples that are closest to the unknown data point, where “k” is a user-specified parameter. These “k” training samples are known as the “nearest neighbors” of the unknown data point.

In classification tasks, the K-NN algorithm uses the class labels of the nearest neighbors to make a prediction for the unknown data point. For example, if the nearest neighbors of an unknown data point are all labeled as “sunny,” then the K-NN algorithm will predict that the unknown data point is also a “sunny.” In regression tasks, the K-NN algorithm uses the values of the nearest neighbors to make a prediction for the unknown data point. For example, if the nearest neighbors of an unknown data point all have a value of “10,” then the K-NN algorithm will predict that the unknown data point also has a value of “10.”

The K-NN algorithm weights the contribution of each nearest neighbor to the prediction. The weight of each neighbor is inversely proportional to the distance between the neighbor and its centroid [39], which is the center of the cluster of nearest neighbors. This means that the closer a neighbor is to the centroid, the more it will contribute to the prediction. In regression tasks, the final prediction is the average of the values of the nearest neighbors.

Several works of literature have used KNN to perform solar PV power forecasting. For instance in [40], the authors explored the impact of classifiers and training data on the accuracy of daily weather classification and employed two commonly used classification methods, KNN and SVM, to classify daily local weather types for DAST solar PV power

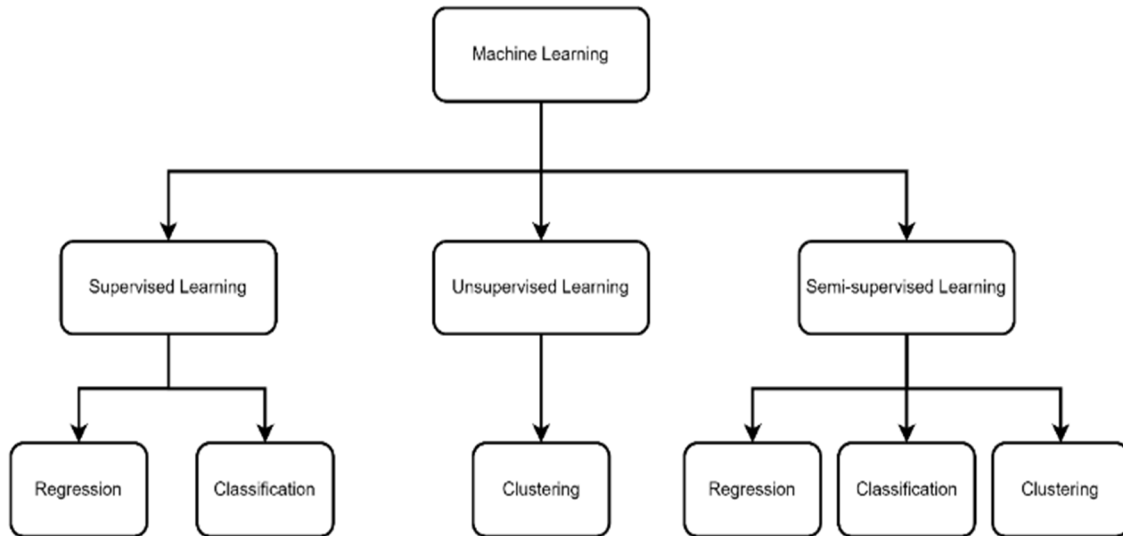


FIGURE 1. Types of ML algorithms.

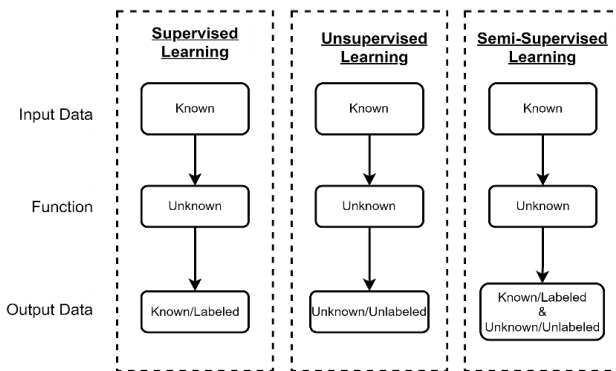


FIGURE 2. ML training process for supervised, unsupervised, and semi-supervised learning.

forecasting. The study found that SVM performs well with small sample scale while KNN is more sensitive to the length of the training dataset and can achieve higher accuracy than SVM with sufficient samples. Overall, this paper provides valuable insights into how weather classification can improve solar PV power forecasting.

b: SUPPORT VECTOR MACHINE (SVM)

Vapnik [41] introduced the support vector machine (SVM) concept which is one of the kernel-based ML techniques designed for classification as well as regression tasks. Support Vector Regression machines (SVR) are the implementation of the SVM concept on regression tasks. SVM operates by use of kernel functions and the Kernel Trick to map low dimensional data into a higher dimensional feature space, based on that linear solutions in the higher dimensional feature space correspond to non-linear solutions in the lower dimension original state. Thus, SVM is a viable option for dealing with a variety of naturally non-linear problems [42].

TABLE 2. Kernel functions Used in SVM models.

Kernel Function	Expression
Linear	$k(x, x') = x^T x'$ (1)
Polynomial	$k(x, x') = (\gamma x^T x' + c)^d$ (2)
Radial basis	$k(x, x') = \exp(-\gamma x - x' ^2)$ (3)
Sigmoid	$k(x, x') = \tanh(\gamma x^T x' + c)$ (4)

A Kernel function denoted as $k(x, x')$, is a symmetrical function that in practice describes the similarity between observations based on their feature values. SVM makes use of several kernel functions which can be categorized as Linear and -Non-Linear hence, by extension, SVMs can be classified, where L-SVM represent SVM that use linear kernel function and K-SVM are for those that use nonlinear kernel functions. The Radial basis function is the most widely used and recommended because of its simplicity in adaptation for optimization, robustness as well as its high efficiency [43]. Examples of kernel functions used in SVMs are provided in TABLE 2.

where x represents the input space and x' represents the training or test samples computed features vector. D represents the degree and c the coefficient.

The mathematical representation of the prediction calculation of an SVM can be represented by Equation 5.

$$y = \sum_{i=1}^N (a_i k(x_i, x') + b) \quad (5)$$

where y is the predicted output value, b is the bias parameter, a_i is the coefficient and x_i is the training set input vector at i th instance with total n training vectors.

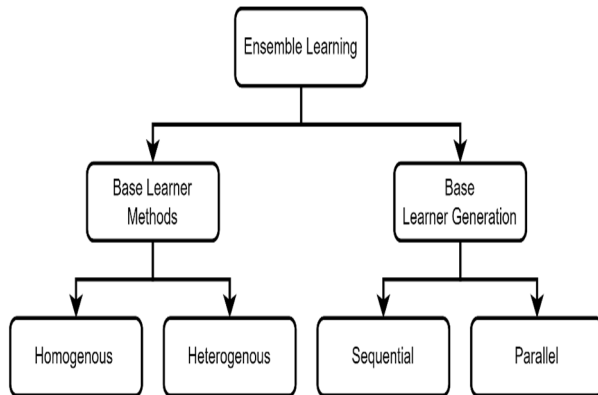


FIGURE 3. Classification of ensemble methods.

SVMs have been used in the works of [44], and [45] to perform solar PV power forecasting. To enhance the accuracy of PV prediction, [46] introduced a PCA-SVM model. This model employed PCA to extract the primary features of the data, which were then used as input for the SVM model. More works were SVMs have also applied to perform solar PV power forecasting include research by authors [47], [48].

c: DECISION TREE AND REGRESSION TREE

A decision tree is a type of ML algorithm that is used to predict the value of a target variable based on the values of other input variables. When the target variable is continuous, meaning that it can take on any real number value, the decision tree is called a regression tree. The algorithm works by dividing the input data recursively into smaller groups based on the values of the input variables and then making predictions about the target variable by averaging the values of the data points in each group. This allows the model to capture complex non-linear relationships in the data and make more accurate predictions.

d: ENSEMBLE

In literature, ensemble learning has been identified as an effective way to increase prediction accuracy by combining and exploiting the advantage of different methods. Ensemble learning is a method of combining the prediction capabilities of multiple models called weak or base learners to improve predictive performance and accuracy by reducing bias and variance exhibited by single-stage models. The key idea is to appropriately combine models, to generate more accurate and resilient models. By combining the efforts of several weak learners, the stability and accuracy are improved as compared to single-stage machine learning algorithms.

Ensemble learning can be described based on the two features, either on the type of base learner or based on how the base learners are connected or generated. Fig. 3 shows an illustration of how ensemble learning methods can be classified. When described according to the base learner, two methods exist: homogenous or heterogeneous. A homogenous ensemble uses a single base learning algorithm to pro-

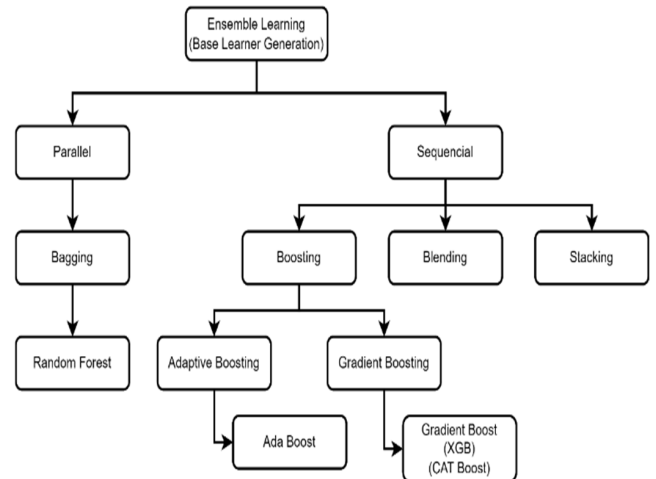


FIGURE 4. Classification of ensemble learning based on learner generation.

duce homogeneous base learners, whereas a heterogeneous ensemble uses different learning algorithms to produce heterogeneous learners. To achieve more accuracy compared to its individual members, ensemble methods require the base learners to be as accurate and diversified as possible.

When classifying according to how the base learners are generated, ensemble learning techniques can further be divided into two groups as shown in Fig. 4. These are sequential ensemble and parallel ensemble. It is important to note that, the ML models used to create the ensembles can homogenous or heterogeneous. In a sequential ensemble, base learners are generated sequentially. The basic motivation of sequential methods is to exploit the dependence between the base learners to improve upon the predictions of the prior base learners. Examples of sequential ensemble methods include boosting, blending, and stacking. For Parallel ensemble, base learners are generated in parallel. The main idea is to exploit the independence and variance between base learners and reduce the variance of the whole model by averaging the errors. A popular method of parallel assemble is bagging.

e: BOOSTING

Boosting is a sequential method used in ML for both regression and classification. Boosting is based on using the advantage of some weak learners and generating them sequentially with each learner improving upon the weakness or poor performance of the previous weak learner. Boosting algorithms can differ in how they create and aggregate weak learners during the sequential process, however, they are similar in how they use errors and weights to improve the performance of the following learners. This is done by monitoring the errors of previous models and using weights based on model errors in the generation of new models or the aggregation of the results of the weak learners. The redistribution of weights helps the algorithm identify the parameters that it needs to

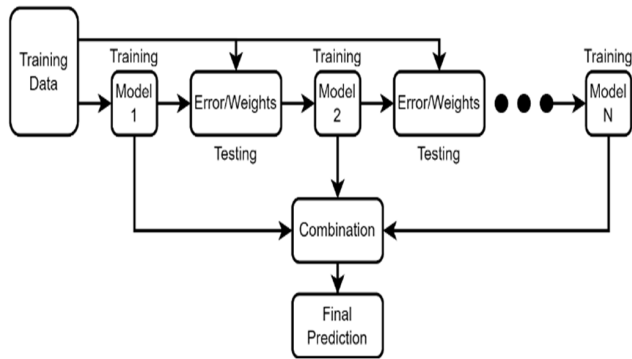


FIGURE 5. Structure of boosting process.

focus on to improve its performance. Fig. 5 provides a depiction of the boosting process, whereas equation 6 describes its mathematical process.

$$f(x) = \sum_t \alpha_t h_t(x) \quad (6)$$

where $f(x)$ is the weighted result forming the strong learner, α_t weight of the weak learner and $h_t(x)$ is the prediction result of a weak learner.

f: BAGGING

Bootstrap aggregating or abbreviated as bagging [49] is a method that combines bootstrap sampling, parallel ensemble learning and aggregating of weak learners to create a model. Bootstrapping or bootstrap sampling is a resampling method that utilizes row sampling with replacement to randomly extracts data into several data sets. It is a method popularly used in statistics to estimate statistical values of small datasets by creating several data samples called bootstrap samples or bootstraps from the original data set. When calculated and combined, the average of statistical values of each bootstrap achieves a better estimate of the population's statistics as compared to direct estimates from the single original dataset [50]. Each generated bootstrap sample data set is then used to train a weak learner, hence multiple weak learners are created to form a parallel ensemble. The model's final output is produced by aggregating the results produced by the ensemble of trained models, where regression models use averaging and classification used voting. Thus, resulting in a small variance for the forecast results. Fig. 6 provides a depiction of the bagging process, whereas equation 7 describes its mathematical process.

$$f(x) = \frac{1}{B} \sum_{b=1}^B f_b(x) \quad (7)$$

where $f(x)$ is the aggregated result forming the strong learner, B is the number of bootstrap samples and $f_b(x)$ is the prediction result of a weak learner.

Examples works from the literature that have utilized bagging in solar PV forecasting include [30], [51], and [52].

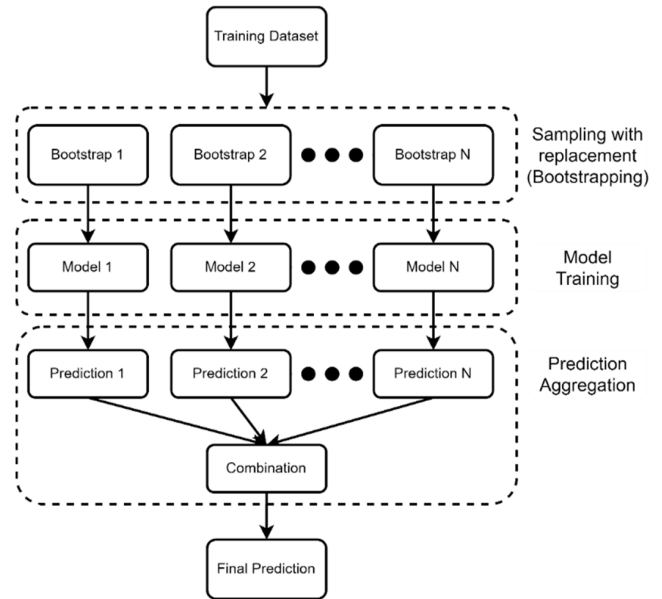


FIGURE 6. Structure of the bagging process.

g: STACKING AND BLENDING

Introduced in 1992, Stacking also known as stacking Generalization [53] is a hybrid of parallel and sequential ensemble, where heterogenous weak learners are trained on the same training data in a parallel manner to produce results or predictions that will be combined with true values to form a dataset that is used as training data to train a meta-model or final estimator. The goal is to reduce the generalization error of prediction models by the use of a meta-model. Thus, the meta-model becomes the strong learner that produces the final result of the prediction model. It is common that 2 training stages or levels are used, where the first stage is taken as level 0 and the next is level 1, however more stages or levels can be added.

Blending is a variation of Stacking, where instead of using the predictions from the whole training dataset or k-fold cross-validation technique to create training data for the meta-model, predictions from a holdout set are used to create training data for the meta-model.

h: RANDOM FORESTS (RF)

Random Forest (RF) is an ensemble machine learning technique that combines multiple decision or regression trees to create a more accurate predictive model. It was developed by the works of [54]. Each tree in a Random Forest contributes to the mean prediction of individual trees, which can help lower the variance of the model. As decision or regression trees are prone to overfitting, which leads to low bias but high variance, Random Forests capitalize on the growth of several trees for each case study, which helps to reduce the variance of the model by averaging the results from each tree. However, the reduction in variance does result in a minor increase in bias while minimizing the variance.

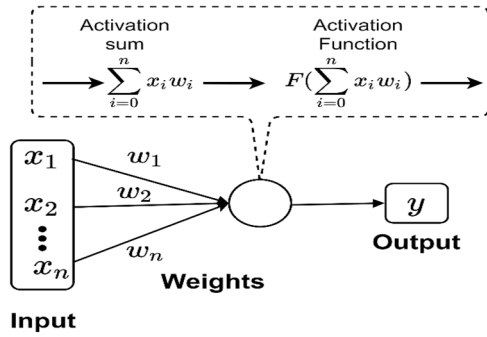


FIGURE 7. Structure of a neuron.

i: ARTIFICIAL NEURAL NETWORKS (ANNs)

An artificial neural network (ANN) is an ML approach modeled on the function and structure of the biological neural network [55]. Artificial neural networks (ANNs) have been used for time series forecasting since the late 1980s [56]. They are comprised of interconnected information-processing units known as neurons [55], [57], [58]. A neuron is also called a node or a perceptron and is a computational unit in which decision-making and information processing occurs. A Neural Network (NN) is formed by the interconnection of the neurons or nodes with connections in which each neuron has one or more weighted input connections, an activation function, and an output connection. Fig. 7 depicts the basic structure of a neuron.

The general architecture of an ANN is separated into three sections or layers: input layer, hidden layer and output layer. Except for input layer neurons, inputs to a given node are multiplied by the weights and then summed to produce a summed activation value. The summed activation is then transformed into the node's specific output or activation through an activation function. Other following nodes progressively process this generated output until it reaches the output layer. Such a method like this has proven to be beneficial in a wide range of scenarios with a big number of inputs. In both linear and nonlinear problems, ANN models can respond well [59], outperforming traditional empirical models [60], [61]. As such ANNs have also been used for solar PV forecasting as seen other works of literature such as [61], [62], and [63].

A variety of topologies exist categorized by the connection forms between input, hidden, and output layers. ANNs can be classified depending on the direction of information flow (Feed-Forward (FF) or Recurrent (R)) or how many hidden layers they have (Single Layer Perceptron (SL) or Multi-Layer Perceptron (MLP)). A brief description of each is given.

An FF-NN is a class of NN where information only flows in one direction. Each neuron in the network is connected such that it only connects to the neurons in the next layer without any loops or feedback in or between the layers. It is one of the most widely used ANNs [64], [65]. Fig. 8 shows the general structure of an FF-NN.

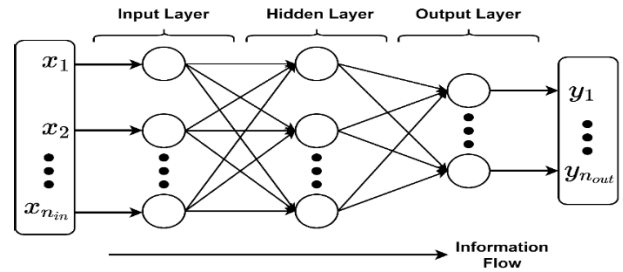


FIGURE 8. Structure of feed forward neural network.

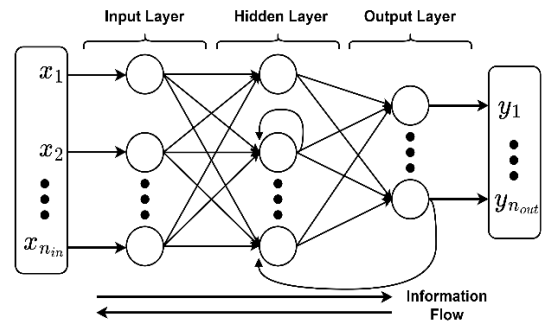


FIGURE 9. Structure of recurrent neural network.

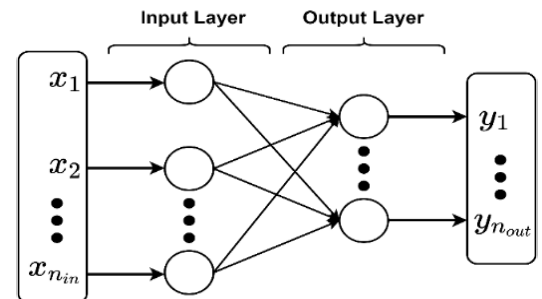


FIGURE 10. Structure of single layer perceptron neural network.

A RNN, is a class of NN structure that introduces a form of memory into a NN through loops or feedback. The feedback is such that the output from a node is kept in the network and used together with the input of the current timestep as input into a node to produce output of the current timestep. Fig. 9 shows an illustration of a RNN.

A Single Layer Perceptron is a NN that does not contain hidden layers, it only has the input and output layer. On the other hand, a Multi-Layer Perceptron is a NN that has one or more hidden layers together with the input and output layers. Fig. 10 and Fig. 11 show an illustration of a SLP and MLP respectively.

j: DEEP NEURAL NETWORK

Focusing on the number of hidden layers in a NN, NNs with less than 3 hidden layers are considered shallow NN or shallow learners whereas NNs with more than 3 hidden layers

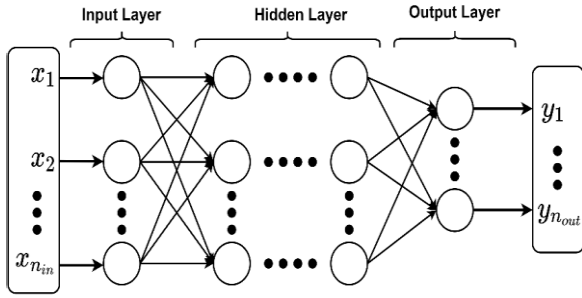


FIGURE 11. Structure of multi-layer perceptron neural network.

are referred to as Deep NN or deep learners. The addition of more layers to a NN adds complexity to the network structure.

IV. PERFORMANCE METRICS

An important aspect of forecasting is model performance and accuracy. Several metrics can provide insight into the accuracy and performance of a model. The use of metrics allows comparisons between different ML models and locations. Each of them emphasizes a different aspect. As a result, the model can never be evaluated by one metric that is valid in all situations. Instead, each of them contributes to measuring the model's accuracy.

In other cases, instead of representing errors as absolute values, normalized performance metrics are used to facilitate comparisons between models or datasets of different scales and sizes. The literature describes several methods for accomplishing performance metric normalization, the majority of which involves dividing the target matrices by statistics based on observational data. For normalized metrics, the letter "N", "n" or "r" is appended to indicate that they have been normalized. Some of the common ways to normalize are:

- 1) Mean (average).
- 2) Standard deviation.
- 3) Range, i.e., difference between the maximum and minimum observations.
- 4) Interquartile range and standard deviation.
- 5) Solar PV system size or estimated capacity.

Below are a group of commonly used performance metrics [5]. The matrices stated below are not unique to any field and can be used to evaluate both PV solar energy harvest prediction and load demand forecasting.

A. NORMALIZED ERROR (nE)

$$nE = \frac{P_m - P_p}{\max(P_{pred})} \quad (8)$$

where; P_p is the predicted power output or load demand; and P_m is the measured power output or load demand.

B. MEAN ABSOLUTE ERROR (MAE)

It is a metric that represents the average magnitude of forecast errors between the actual or measured values and the

forecasted or model predictions. It is suitable for evaluating uniform forecast errors.

$$MAE = \frac{1}{N} \sum_{i=1}^N |P_m - P_p| \quad (9)$$

C. MEAN BIAS ERROR (MBE)

Captures a forecasting model's systematic tendency to overestimate or underestimate.

$$MBE = \frac{1}{N} \sum_{i=1}^N (P_m - P_p) \quad (10)$$

D. ROOT MEAN SQUARE ERROR (RMSE)

Root mean squared error (RMSE) is a measure of how well a model's predictions match the true values. RMSE is calculated by taking the difference between the true values and the predicted values, squaring this difference, and then averaging the squared differences. By squaring the differences, RMSE gives larger errors a higher weight, which means that outliers will have a greater impact on the RMSE value. This means that models with a lower RMSE are generally considered to be more accurate because they make fewer large errors.

$$RMSE = \sqrt{\frac{1}{N} \sum_{i=1}^N (P_m - P_p)^2} \quad (11)$$

E. NORMALIZED OR RELATIVE ROOT MEAN SQUARE ERROR (nRMSE/rRMSE)

$$nRMSE = \frac{\sqrt{\sum_{i=1}^N (P_m - P_p)^2}}{P_c} \quad (12)$$

where; P_c is the solar PV system power output capacity.

F. MEAN ABSOLUTE PERCENTAGE ERROR (MAPE)

Mean absolute percentage error (MAPE) is a measure of forecast accuracy that is similar to mean absolute error (MAE). Both measures are used to evaluate the error of a model's predictions, but MAPE is expressed as a percentage rather than as an absolute value. MAPE is calculated by taking the absolute value of the difference between the true value and the predicted value and then dividing this value by the true value. This percentage error is then averaged over all the predictions made by the model. Like MAE, MAPE is best used when the errors of the model are uniform or consistent across all predictions.

$$MAPE = \frac{100}{N} \sum_{i=0}^N \left| \frac{P_m - P_p}{P_c} \right| \quad (13)$$

where; P_c is the solar PV system power output capacity.

G. MEDIAN ABSOLUTE PERCENTAGE ERROR (MdAPE)

Median absolute percentage error (MdAPE) is calculated by taking the median of the absolute percentage errors between

the true values and the predicted values. MdAPE is similar to MAPE, but it is less sensitive to outliers because it uses the median instead of the mean to average the errors. MdAPE is often used as a measure of forecast accuracy, particularly when the true values and predicted values have different scales or units of measurement.

$$MdAPE = median \left| 100 \frac{P_m - P_p}{P_m} \right| \quad (14)$$

H. SKILL SCORE (SS)/FORECASTING SKILL

Skill Score is used to compare one forecasting model with another. It can be approximated as [5];

$$SS_{model}^{RMSE} \approx 1 - \frac{RMSE_{model}}{RMSE_{baseline}} \quad (15)$$

When using a persistence forecast as a reference or baseline model, the SS score of the persistence model against itself is equal to 0. A model that exhibits better forecasting accuracy when compared to the reference model results in an SS greater than 0. The higher and closer to 1 the SS, the more accurate and better performing the forecasting model is. However, models with SS less than 0, indicate a decline in performance and lower accuracy of the model being tested.

I. COEFFICIENT OF DETERMINATION (R^2)

An R^2 is a measure of how much of the variability in a data set can be explained by a regression or prediction model. For predictions that are no different to guessing, the R^2 score would be close to zero; indicating that much is unexplained by the regression model. Values close to one mean that the regression or prediction model can capture and or predict the dataset well.

$$R^2 = 1 - \frac{\sum_{i=1}^n (P_m - P_p)^2}{\sum_{i=1}^n (P_m - \bar{P}_m)^2} \quad 0 \leq R^2 \leq 1 \quad (16)$$

V. COMPARISON OF ML PV POWER FORECASTING METHODS AND CORRESPONDING RESEARCHERS

In literature, many techniques have been studied and used to perform solar PV power forecasting. Each method has advantages and disadvantages and often, the performances of the forecasting methods are comparable. However, depending on the available data, each problem might be solved with a different approach and undergo different optimizations, which would result in other PV power forecasting methods significantly outperforming other methods. Therefore, there is merit in testing and reviewing the different forecasting approaches found in the literature. This section provides an overview of related works primarily comprised of TABLE 4, which shows some works from 2010 – 2022 in PV power forecasting.

A. ANALYSIS OF OF ML PV POWER FORECASTING METHODS

When evaluating the performance of solar PV forecasting models, it is important to use accurate and consistent performance metrics that enable fair comparisons between different

TABLE 3. The difference between parallel and sequential ensemble learning.

	Bagging	Boosting
Processing Method	Parallel	Sequential
Overfitting to Dataset	No	Possible
Selection of Training Dataset	Training samples are randomly selected	Training samples for each training instance are fixed
Optimization	Easy	Difficult
Sample Weights	Uniform Sampling, hence, each sample has equal weights	Sample weights are adjusted based on error rate. Higher errors lead to higher sample weights.
Prediction Functions	Prediction Functions have the same weights.	Weights are adjusted based on error rate. Smaller errors lead to greater weights for the prediction function.
ML Model Examples	Random Forest	CAT Boost, XGBoost, LightGBM, GBDT

models. The RMSE and MBE are commonly used metrics but comparing the performance of different ML models can be challenging when they operate on different scales of PV systems with varying maximum theoretical production capacity. To address this issue, normalized or relative error metrics, such as the normalized NRMS, are recommended. However, few papers from the literature as shown in the comparative table present the accuracy of their models in a normalized or relative error form, making it difficult to compare the forecasting performance of different ML models. One potential solution to this challenge is to normalize error metrics such as RMSE and MBE using the maximum theoretical production capacity of the PV system, which enables a fair comparison of different ML solar PV forecasting models by removing the bias caused by the varying maximum theoretical production capacity of the PV systems. Using a normalized error metric or normalizing RMSE and MBE can help to identify the best performing models for different scales of PV systems. However, some cases arises where some information are not available for some papers, and we indicate such situation using “NA”.

Based on TABLE 4, the solar PV forecasting model with the least NRMSE is that by [66], followed by the works of [67]. Both methods realized the models by utilizing the hybrid or ensemble approach, where [67] combined ANN and post-processing with a linear regressive model thereby making a hybrid or series ensemble with NRMSE of 7.53% in a cold climate. While [66] using parallel ensembles, introduced a

TABLE 4. ML PV power forecasting methods.

Authors/year	Forecast horizon	Forecast resolution	Time Period	Location	Data Pre-Processing	Method	Variables	Single Point Forecast Normalized Evaluation
Ramsami and Oree [68]	Day-ahead	60 min (1h)	03/02/2012-30/12/2013	Unite Kingdom, Cornwall, Newquay	Feature selection (Stepwise Regression) Delete Missing Records	Input optimized hybrid ANN via stepwise regression with NWP variables	The weather forecasted data (GHI, wind speed, sunshine duration, relative humidity, temperature, pressure, rainfall and wind direction)	NA
Mohammed <i>et al.</i> [69]	1h – Day-ahead	60 min (1h)	01/04/2012-31/05/2014	NA	NA	Probabilistic forecasting based on an ensemble of statistical techniques	Historical PV power and NWP variables from the ECMWF	NA
Lu <i>et al.</i> [70]	Day-ahead	1 Hour	23/01/2014-20/08/2014	United States, Marana, Arizona	NA	GHI-P PV ML model by a combination of generated inputs from 3 NWP models.	Historical values of GHI and Weather forecasted data from several NWP models	NA
Lin and Pai [71]	1 month	1 month	January 2010 to April 2014	Taiwan	Seasonal decomposition	Evolutionary seasonal decomposition LS-SVR for monthly prediction of the PV power output in Taiwan	Historical PV power	MAPE 7.84%
Huang and Perry [72]	1–24 h	60 min (1h)	01/04/2012-31/05/2014	NA	low-pass filters using the Fourier transformation for modelling the annual trends	Deterministic forecast based on Gradient boosting and probabilistic forecast based on k-NN regression.	Onsite and neighboring historical PV power and weather forecasted data from ECMWF	NA
Alessandrini <i>et al.</i> [73]	0–72 h	60 min (1h)	January 2010 to December 2011 (Catania) July 2010 to December 2011 (Milano) April 2011 to March 2013 (Calabria)	Italy (Milano, Catania and Calabria)	NA	Analog Ensemble and QR, providing a likely set of Power	Historical PV power, PV system Properties (solar azimuth and elevation angles) and Weather forecasted data (GHI, temperature, cloud cover)	NMAE(Nominal) 6–7%
Almeida <i>et al.</i> [74]	Day ahead	60 min (1h)	January 2009 and December 2010	Spain	NA	QRF considering Spatio-temporal variability indexes and different training sets	Historical PV power, IPOA, temperature, Wind speed and weather forecasted data.	NA
De Felice <i>et al.</i> [75]	1–10 days	NA	(18 and 24 months)	Italy (65 Sites)	NA	SVR	Estimations of GHI and Temperature and Weather forecasted data (GHI, temperature)	NA
Zamo <i>et al.</i> [76]	28–45 h	60 min (1h)	2009 to 2011	France	NA	Benchmark: Generalized additive models, SVM, LM, binary regression tree and ensemble trees (RF, boosting, bagging)	Weather forecasted data from different NWP models.	NRMSE (Maximum measured Production) 10%-12%
Li <i>et al.</i> [77]	Day-ahead	24h (1 day)	January 1 2011 To June 30, 2012	Coloane Islands, Macau	NA	Benchmarks: ANN and Time series models (ARIMA, ARMAX, simple and double moving average, etc.)	Endogenous: Historical PV power Exogenous: Weather forecasted data (Sunshine hours, temperature(max and min), wind speed, relative humidity, wind direction and dew temperature)	NA
Long <i>et al.</i> [78]	Intraday up to 3 days	24h (1 day)	July 2010 to July 2012	Coloane Islands, Macau	Parameter selection	Benchmarks: K-NN, MLR, SVM and ANN	Historical PV power, Wind speed, GHI, rainfall, sunshine hours, relative humidity and dew temperature	NA
Mellit <i>et al.</i> [79]	Day-ahead	60 min (1h)	January 1 2011 to December 31 2011	Puglia, Italy	NA	ANN using classification of days based on forecasted GHI.	Historical PV power, GHI and cell temperature and Weather forecasted data temperature and GHI	NA

TABLE 4. (Continued.) ML PV power forecasting methods.

Zamo <i>et al.</i> [80]	66–72 h	60 min (1h)	2009 to 2011	France	NA	Evaluation of the utilization of a single or multiple NWP models and use of probabilistic predictions using LM QR and QR Forest.	Weather forecasted data from different NWP models	NA
Gao <i>et al.</i> [81]	Day-ahead	60 min (1h)	Nov 1, 2016 to Oct 28, 2017	Shandong province, China.	Classification of days based Weather and season clustering	LSTM	Historical PV and meteorology data	Non Ideal (Rainy days) MAE% 3.37%
Theocharides <i>et al.</i> [67]	Day-ahead	60 min (1h)	NA	Nicosia, Cyprus and Albuquerque, New Mexico (NM), USA	Weather clustering (K-means)	ANN + Post processing (linear regressive correction)	Historical PV power and numerical weather prediction data	NRMSE 6.11% (hot climate) NRMSE 7.35% (cold climate)
Pretto <i>et al.</i> [66]	Day-ahead	60 min (1h)	2017 to 2019 (3 years)	Politecnico di Milano, Italy	clustered according to the solar irradiation forecast and Clearness Index.	Probabilistic Ensemble Method for an Enhanced Day-Ahead PV Power Forecast [ANN]	Historical PV power and numerical weather prediction data, Clear Sky condition	Cloudy day NRMSE 4.79%
Nespoli <i>et al.</i> [63]	Day-ahead	60 min (1h)	2017	Politecnico di Milano, Italy	Enhanced Normalization	Selective MLP ANN Ensembles	Historical PV power and numerical weather prediction data, Clear Sky condition	NRMSE 17.62%
Abdellatif <i>et al.</i> [82]	Day-ahead	NA	1 January 2018 and 31 December 2021	Malaysia	Normalized based on standard deviation.	Stacked ensemble algorithm. Baseline [RF,XGBoost,AdaBoost] Meta [extra trees regressor (ETR)]	Historical PV power, Time, solar irradiance, wind speed, ambient temperature, PV module temperature,	NA
Khan <i>et al.</i> [83] Case Study 2	1h	1h	4 years	Breda Netherlands	Data Cleaning and scaling	Stacked ensemble algorithm Baseline [ANN,LSTM] Meta[XGBoos]	Historical PV power and numerical weather prediction data (Historical PV power and numerical weather prediction data)	NA
Pombo <i>et al.</i> [14]	1-3 Days-ahead	1h	June 2019 to the 31 August 2020 (15 Months)	Roskilde, Denmark	Data Cleaning and scaling	Physics informed ML-models	Historical PV power and local weather data	NA
Essam <i>et al.</i> [15]	5 min	5 min	21st January 2011 to 3rd March 2012 (1 year)	United States [subtropical-Cocoa, Florida; marine west coast-Eugene, Oregon; and semi-arid-Golden, Colorado]	Data Cleaning, scaling and feature selection	ANN, RF, DT, XGB, and LSTM	weather parameters and solar power output	NA

new ensemble, called probabilistic ensemble method (PEM), for improving the forecasting performances in cloudy days, hence resulting in a NRMSE of 4.79%.

Evaluating the forecasting performance of solar PV models in adverse weather conditions is essential to showcase their robustness and versatility. Atmospheric conditions significantly impact the performance of the solar PV forecasting models, and variations can pose significant challenges to the accuracy of the models. As shown in the comparative table, mostly recent literature included performance analysis of solar PV forecasting models during adverse weather conditions to assess their reliability in providing accurate forecasts. By including such information, researchers can

better understand how their models perform under different environmental conditions and identify the best models for various weather conditions and seasonal changes.

Geographic location is a crucial factor in evaluating the performance of solar PV forecasting models. Few works of literature as shown in the comparative table provided information on the geographic location of the PV systems used in their studies, and this can significantly influence the performance of the solar PV systems. Models trained on data from PV systems in different locations would have varying degrees of accuracy when applied to other locations. Therefore, it is essential to take into consideration the geographic location of the PV systems used when evaluating and comparing

the performance of solar PV forecasting models. Evaluating models based on their performance in the same geographic location or at least in regions with similar climatic conditions is necessary to obtain accurate and meaningful results.

B. ML EXPERIMENT SIMULATIONS

To further enhance the comparison and provide more insights into the advancement in the area, we simulate the performance of different ML methods used in solar PV power forecasting. The ML models selected to be simulated include Support Vector Regressor (SVR), SVM, Random Forest (RF), Gradient Boosting (GB), Extreme Gradient Boost (XGB), Multi-Layer Perceptron and a lasso regressor. The models were selected based on their effectiveness in regression tasks and their suitability for handling large datasets with high dimensionality. Additionally, we considered the ability of the models to capture the complex non-linear relationships between the input features and the target variable in solar PV power forecasting. Based on these criteria, we selected a range of ML algorithms that have been widely used in solar PV power forecasting and have shown promising results in previous studies. Section VI detailed our experimental simulations and analysis of the different methods.

VI. EXPERIMENT SETUP

A. EXPERIMENT HARDWARE

The experiment is implemented in Windows 10 64-bit Desktop Computer with an Intel(R) Core (TM) i7-10750H CPU @ 2.60 GHz processor, 1TB HDD and 8 GB DDR3 RAM. The code was written in Python, and ran in a Jupiter notebook environment, with Microsoft Visual Studio Code used as an IDE.

B. EXPERIMENT SOFTWARE PACKAGES

The main software package used in the experiment is the Scikit Learn library, with NumPy and Pandas as support packages. Scikit Learn is an open-source python library that is a simple and efficient tool for statistical modeling, predictive data analysis, and machine learning. It is based on python packages NumPy, SciPy, and Matplotlib. It provides efficient methods for dealing with common machine-learning tasks such as classification, regression, clustering, model selection, and dimensionality reduction. Furthermore, Scikit learn can be used to create models for both supervised learning and unsupervised learning.

C. EXPERIMENT DATA

Data used in this work consist of two datasets, these being PV power production time series and forecasted weather time series. Further information regarding each dataset is given.

1) ROOFTOP SOLAR PV SYSTEM POWER GENERATION DATA

This work used a dataset that is an extract from an open-access dataset by Visser et al.; published in the Journal of

TABLE 5. Solar PV power production dataset information.

Resource Metadata	
Data Source:	https://explore.openaire.eu/search/dataset?pid=10.5281%2Fzenodo.6906504
Data:	Residential Rooftop PV power production
Location:	City: Utrecht, Country: Netherlands
Dataset period:	2014 - 2017 (4 Years)
Time	1 minute
Resolution:	
Number of Variables:	175
Number of Records:	2075259
Missing Records Percentage:	1.25%

Renewable and Sustainable Energy [84]. It contains an extensive 175 time series of rooftop-mounted PV system power measurements of residential located in the city of Utrecht in the Netherlands from the year 2014 to 2017. Each rooftop PV production time series is recorded at 1-minute intervals and consists of 4 years of data from December 2014 to November 2017. Information about the solar PV dataset is summarized in TABLE 5.

Apart from the PV power production records, the open access dataset also includes a metadata file of the PV systems, including each PV system's unique ID, estimated DC and AC capacity, annual yield, start and end time of registered power measurements, azimuth and tilt angle, and mapped grids of the system location. The dataset has been used by several works in literature to carry out predictions [85], [86], [87].

2) DAY-AHEAD FORECASTED WEATHER DATA

The weather data used in this work is publicly available from European Centre for Medium-Range Weather Forecasts (ECMWF). It consists of 24 hours day ahead, 1-hour resolution weather forecasts data corresponding to the same time and location as the solar PV dataset. The weather data has a total of 14 weather-related parameters and 2 date and time information parameters, hence a total of 16 variables are included in the dataset. Information about the meteorological dataset is summarized in TABLE 6.

D. EXPERIMENT METHODOLOGY

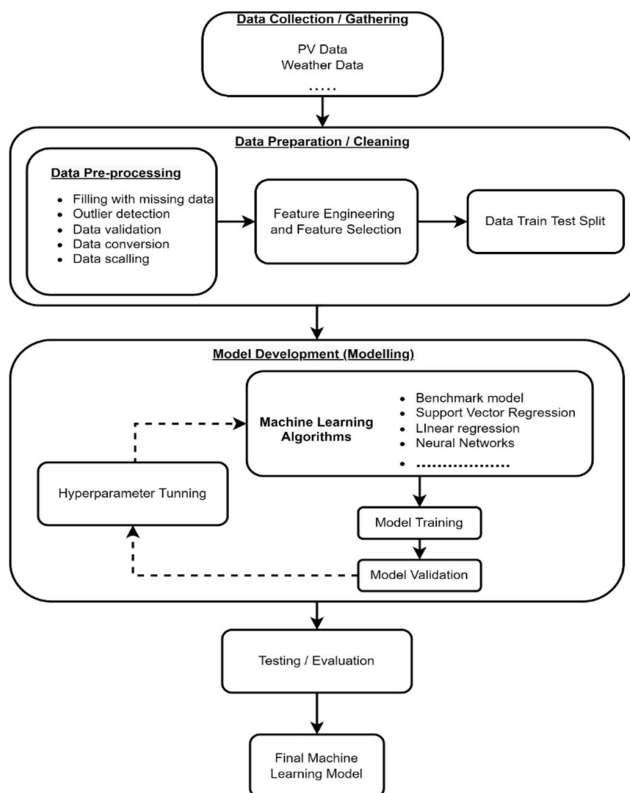
The purpose of this experiment is to analyze the dataset and explore the application of machine learning (ML) methods to develop regression models to make predictions of residential Rooftop PV power production. Implementation of the work is carried out using the framework as shown in Fig. 12.

TABLE 6. Meteorological dataset information.

Resource Metadata	
Data Source	European Centre for Medium-Range Weather Forecasts (ECMWF)
Data	Day-Ahead Forecasted Weather data
Location	City: Utrecht, Country: Netherlands
Dataset period	2014 - 2017 (4 Years)
Time	1 hour
Resolution	
Number of Variables	16
Number of Records	35065
Missing Records Percentage	0%

TABLE 7. Feature variables for model creation.

Data Type	Feature Variable	Units
Calendar Information	Hour of Day	-
	Day of Month	-
	Day of Year	-
	Month	-
	Year	-
Historical Solar PV Power Generation	PV Power (24H lag)	W
	PV Power(48H lag)	W
	PV Power (72H lag)	W
Day Ahead(+24H) NWP information	Ambient temperature	C
	Dew temperature	C
	High Cloud cover	Range (0, 1)
	Medium cloud cover	Range (0, 1)
	Low Cloud cover	Range (0, 1)
	Total cloud cover	Range (0, 1)
	Total precipitation	m
	Mean Sea-level pressure	PA
	Surface pressure	PA
	Surface Solar Radiation (SSR)	W/m ²
	Surface Solar Radiation Downward (SSRD)	W/m ²
	Eastward Wind Component @10m (u ₁₀)	m/s
	Northward Wind Component @10m (v ₁₀)	m/s
	Wind Speed	m/s

**FIGURE 12.** Machine learning solar PV power production framework.

1) DATA PREPARATION

Machine learning algorithms build prediction models based on data; hence data preparation and cleaning are essential steps in building ML forecasting models. The use of unprocessed data leads to a loss in the forecasting model accuracy. The weather data was clean and had no missing records. The solar PV power dataset underwent multiple processes where it is analyzed and cleaned. This involved filling in missing data and identifying outliers.

Furthermore, additional processes such as feature engineering and feature selection may be performed on the dataset to improve it. The goal of the experiment is to design a predictive model to perform solar PV power prediction on 24h ahead time horizon. Hence, the target variable was set to be solar PV power production while weather data and time properties were used as feature variables. The data underwent feature engineering, where additional features were derived from the existing data and the domain knowledge of solar PV power production. For feature engineering, information was extracted from the date and time features, the extracted data included the day of the month and day of the year. Three extra features were generated from the solar PV data by adding 24, 48 and 72 hours of lagged data. TABLE 7 shows the complete list of the feature variables used in the ML model development stage.

TABLE 8. Grid search hyperparameter search space.

Model	Hyperparameters	Grid Search	Selected Parameters
Lasso	Alpha	[0.006, 0.008, 0.009, 0.01, 0.015, 0.017, 0.019, 0.02]	0.02
	Fit-Intercept	[1, 0]	1
MLP	Alpha	[1e-5, 1e-4, 1e-3, 1e-2, 1e-1, 0.15, 0.18, 0.2, 0.25, 0.3]	0.2
	Activation	['relu', 'identity', 'tanh']	relu
	Hidden Layer Size	[(10,2), (10,3), (10,4), (10,5), (10,6), (10,8)]	(10,4)
	Max Iterations		2000
	Optimization		Adam
SVR	C	[27, 28, 29, 30, 31, 32, 33]	27
	Epsilon	[14, 15, 16, 17, 18, 19, 20]	14
	Fit-Intercept	[1, 0]	1
SVM	C	[0.2, 0.4, 0.5, 0.6, 0.7, 0.8, 0.9]	0.9
	Gamma	['scale', 'auto', 0.1, 0.2, 0.3, 0.4]	scale
	Kernel	['rbf', 'sigmoid', 'poly']	sigmoid
RF	Max Features	['log2', 'sqrt']	log2
	Min Samples Split	[5, 6, 7, 8, 9, 10, 11]	7
	N Estimators	Range (100, 350, 10)	100
XGB	Eta	[0.006, 0.008, 0.009, 0.01, 0.015, 0.017, 0.019, 0.02]	0.02
	Gamma	Range (150, 310, 10)	260
	N Estimators	Range (70, 90, 2)	88
	Subsample	[0.5, 0.75, 1]	0.5
GB	Learning Rate	[0.005, 0.007, 0.009, 0.01, 0.02, 0.03]	0.005
	Max Depth	[2, 4, 6]	4
	Min Samples Split	[5, 6, 7, 8, 9, 10, 11]	10
	N Estimators	Range(400, 500, 10)	410

The holdout validation process was used to train and test the predictive models, hence the data employed in this study was divided into two subsets. The training set contained 70% of the data and the test set included the remaining 30%. A larger training set was used to ensure the reliability of the predictive model.

2) MODEL DEVELOPMENT

In the model development, several ML algorithms were used to train and create the prediction models. The ML prediction models created in the work include Gradient Boost (G Boost), extreme gradient Boost (XG boost), Support Vector Regression (SVR), Random Forest (RF), Lasso Regression (LR), Ada Boost and A Multi-Layer Perceptron neural network (MLP NN). The default parameters of regression models in the Scikit Learn library were used as a basis for the hyperparameters of each model. The optimized models' hyperparameters were chosen by conducting a grid search optimization within the range indicated in TABLE 8. The search area was determined by building upon the default hyperparameters of Scikit Learn.

When evaluating the performance of forecasting procedures, baseline (benchmark) methods are used to compare. Two methods were used to produce the baseline test results. The baseline methods used in the experiment were based on assumptions of persistence about the underlying process of solar PV power production. This includes

24 hours Persistence method and the time of day daily average method.

a: 24H PERSISTENCE

The 24H persistence method uses the previous day's observed power output as the forecasted value for solar PV output. This is expressed in the equation.

$$P_p(t) = P_m(t - 24) \quad (17)$$

b: TIME OF DAY DAILY AVERAGE (AVERAGE)

This method used the average of previous solar PV power production based on time as the forecast of the next day at the said time of day.

To improve the performance of the ML models, hyperparameter optimization was carried out on all the models. Grid search hyperparameter optimization was used as the optimization method.

3) TESTING/EVALUATION

Several evaluation metrics are used to compare the performance of the developed forecasting models. Each metric emphasizes a different aspect of forecasting hence using several evaluation metrics provides insight into the accuracy and performance of a model. The forecast evaluation metrics used in this work are shown in TABLE 9.

TABLE 9. Forecast evaluation metrics.

Metrics	Abbreviation	Formula	Range	Goal
Mean Absolute Error	MAE	$= \frac{1}{N} \sum_{i=1}^N P_m - P_p $	$[0, \infty)$	Minimize
Mean Squared Error	MSE	$= \frac{1}{N} \sum_{i=1}^N P_m - P_p^2$	$[0, \infty)$	Minimize
Root Mean Squared Error	RMSE	$= \sqrt{\frac{1}{N} \sum_{i=1}^N P_m - P_p^2}$	$[0, \infty)$	Minimize
Normalized Root Mean Squared Error	nRMSE	$= \frac{\sqrt{\frac{1}{N} \sum_{i=1}^N P_m - P_p^2}}{P_c}$	$[0, \infty)$	Minimize
Coefficient of Determination	R ²	$= 1 - \frac{\sum_{i=1}^n (P_m - P_p)^2}{\sum_{i=1}^n (P_m - \bar{P}_m)^2}$	$(-\infty, 1]$	Maximize
SKILL SCORE	SS	$SS_{model}^{RMSE} \approx 1 - \frac{RMSE_{model}}{RMSE_{baseline}}$	$(-\infty, 1]$	Maximize

VII. RESULTS AND DISCUSSION

This section presents the results and discussion of the experiments.

A. BASELINE FORECASTS PERFORMANCE

The baseline methods created made simple forecasts based on strong assumptions of slow change or persistence of daily solar PV power production, hence the performance of the models is not expected to be high. This is evident by the results shown in TABLE 9, while Fig. 13 which shows the nRMSE in percentage, Coefficient of determination and skill score bar graph representation of the model performances.

Between the two baseline methods, the 24hr persistence method displays the least accuracy and prediction performance across all the performance metrics. Such results are to be expected since it only uses the previous day's solar PV power output as the forecast, however, consecutive days can experience vastly different weather conditions hence resulting in large differences in solar PV power output. Furthermore, since the 24hr persistence model only considers one data point instance, having a few data points sampled results in poor forecasting performance. The average (BSL_Average) method shows better performance than the 24hr persistence and this is also expected since it considers more data points; as such it can better reflect the underlying process, hence the high R² above 0.5, which indicates that it is able to model close to half of the underlying process for solar PV power production.

The 24hr persistence method is selected as the reference model when calculating skill score (SS), thus resulting in the SS score of 0 for the 24hr persistence. To make the SS score easier to read, it has been represented as a percentage. The average baseline method shows an SS score of 16.6% which indicates a 16.6% improvement in forecasting accuracy when compared to the 24hr persistence reference model.

Further analysis of the results of the two baseline models through graphical representations is shown in Fig. 14 and Fig. 15. Where Fig. 14 shows a line plot of the residuals across the test data of the two-baseline methods Fig. 15 shows a localized perspective of the model's performance through box plots of the model's absolute relative error for each hour of the day. From Fig. 14 the superiority of the average baseline method over the persistence model is further evident as shown by the lesser residuals in most of the instances in the test data. Fig. 15 further shows that the daily prediction error for the persistence model is more than the average baseline model, hence the higher median values, longer interquartile range, and longer error ranges. Thus, the average model consistently outperforms the persistence model across all the performance metrics as shown in TABLE 10.

B. ML MODEL FORECASTS PERFORMANCE

The results of the ML model performance are shown in TABLE 11 and TABLE 12 shows the results of the ML models that were designed using the default Scikit Learn library model Hyperparameters, whereas TABLE 12 shows

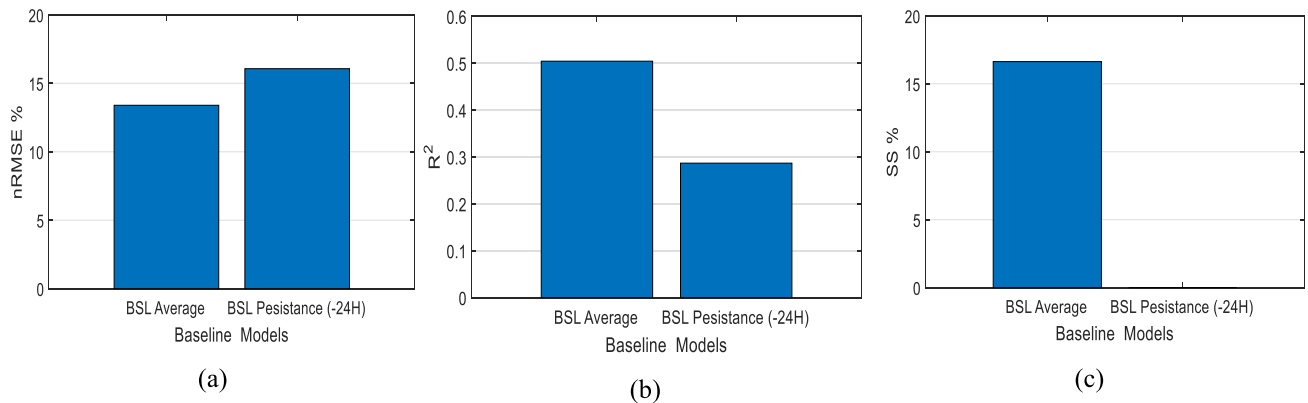


FIGURE 13. Performance of baseline methods on testing data. (a) Normalized RMSE as percentage. (b) Coefficient of determination. (c) Skill score as percentage.

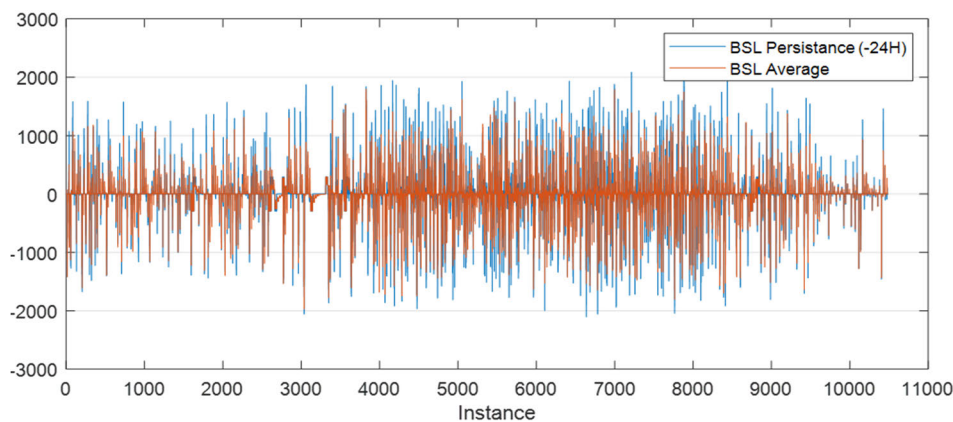


FIGURE 14. Residuals of persistence and average baseline forecast methods.

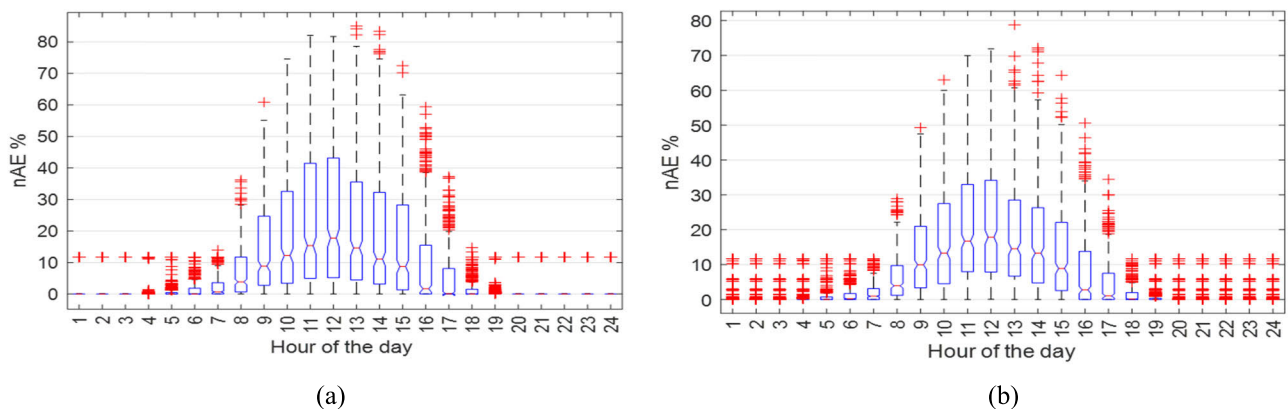


FIGURE 15. Boxplot for absolute relative error for each hour of the day. (a) Results for persistence. (b) Results for average baseline forecast methods.

the performance of the models that were designed with hyperparameter optimization using grid search.

1) ML MODEL WITH DEFAULT HYPERPARAMETER

Based on TABLE 11, it can be noted that the ML models perform better than the baseline methods at forecasting. This

is evident by both the R^2 and SS for all ML models being greater than that of the baseline methods. Furthermore, the superiority of the ML models over the baseline forecasting models is also evident by all other error metrics showing better scores for all the ML models when compared to both the average and persistence baseline methods.

TABLE 10. Performance of baseline forecasts methods.

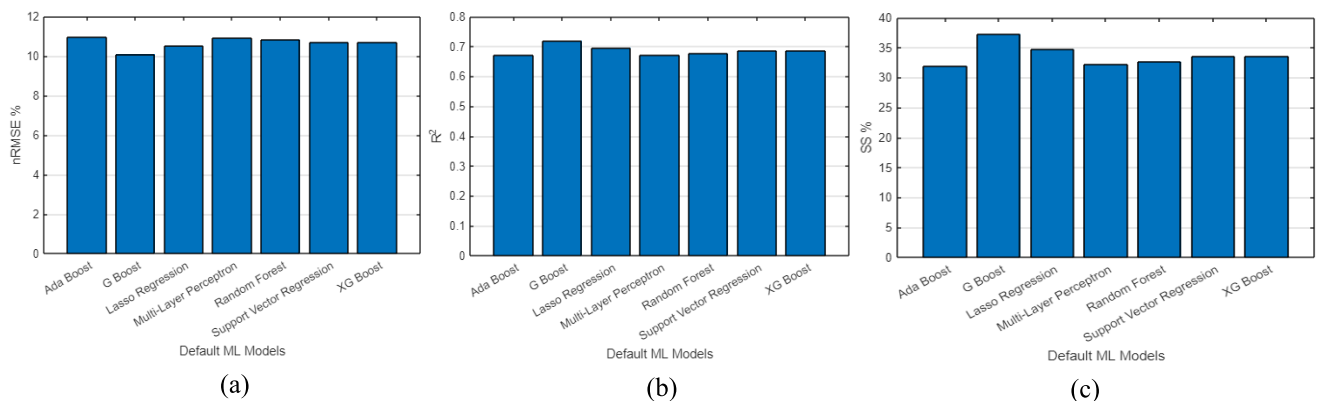
Model	MAE (W)	MSE (W)	RMSE (W)	nRMSE (%)	R ²	SS (%)
BSL Average	160.8677	112458.6305	335.3485	13.39786256	0.5043	16.63557
BSL Persistence (24H)	179.8994	161819.5869	402.2681	16.07143827	0.2868	0

TABLE 11. Performance of ML models with default hyperparameter.

Model	MAE (W)	MSE (W)	RMSE (W)	nRMSE (%)	R ²	SS (%)
G Boost	132.764	63705.257	252.399	10.084	0.719	37.256
Lasso Regression	152.486	69012.693	262.703	10.496	0.696	34.695
XG Boost	143.432	71349.363	267.113	10.672	0.686	33.598
Support Vector Regression	161.473	71411.071	267.229	10.676	0.685	33.570
Random Forest	156.464	73448.297	271.014	10.828	0.676	32.629
Multi-Layer Perceptron	149.804	74386.843	272.740	10.897	0.672	32.200
Ada Boost	151.825	74933.314	273.740	10.936	0.670	31.951

TABLE 12. Performance of ML models after grid search hyperparameter optimization.

Model	MAE (W)	MSE (W)	RMSE (W)	nRMSE (%)	R ²	SS (%)
Random Forest	143.805	63558.398	252.108	10.072	0.720	37.328
XG Boost	136.047	66875.344	258.603	10.332	0.705	35.714
Lasso Regression	157.673	70375.197	265.283	10.599	0.690	34.053
Support Vector Regression	163.895	71868.669	268.083	10.710	0.683	33.357
G Boost	155.420	73714.200	271.504	10.847	0.675	32.507
Ada Boost	154.019	74843.305	273.575	10.930	0.670	31.992
Multi-Layer Perceptron	204.114	86565.513	294.220	11.755	0.619	26.860

**FIGURE 16.** Performance of default hyperparameter ML methods on testing data. (a) Normalized RMSE as percentage. (b) Coefficient of determination. (c) Skill score as percentage.

Focusing on comparing the performance of the ML models with default hyperparameters, as shown in TABLE 11 and Fig. 16, the gradient boost forecasting model is the model with the least forecasting error and highest SS improvement, hence making it the best model. However, the difference

between the best model and all the other ML models is not as significant as compared to the large leap between the ML and the baseline models. The least accurate model in the ML models is the Ada Boost, with an SS difference of 5.3% when compared to the best-performing model.

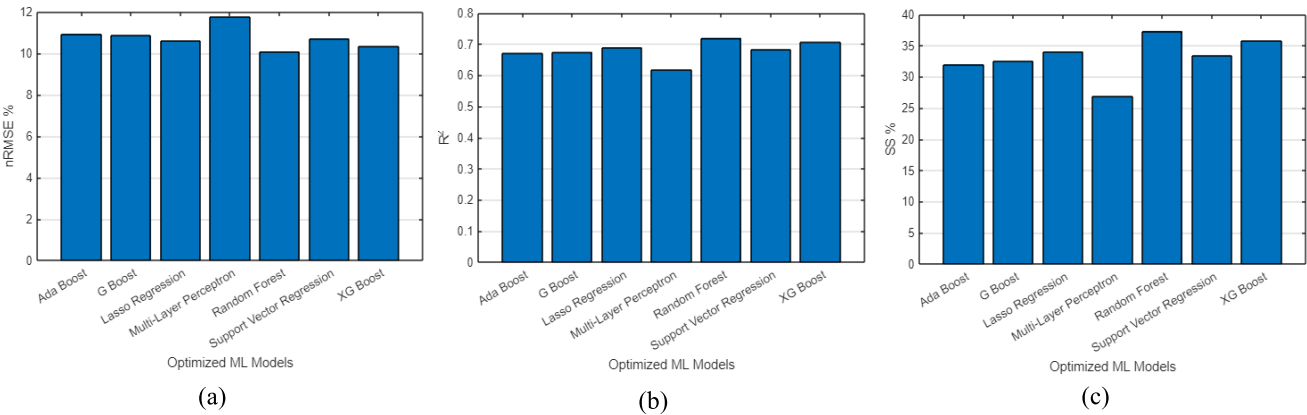


FIGURE 17. Performance of grid search hyperparameter optimization ML methods on testing data. (a) Normalized RMSE as percentage. (b) Coefficient of determination. (c) Skill score as percentage.

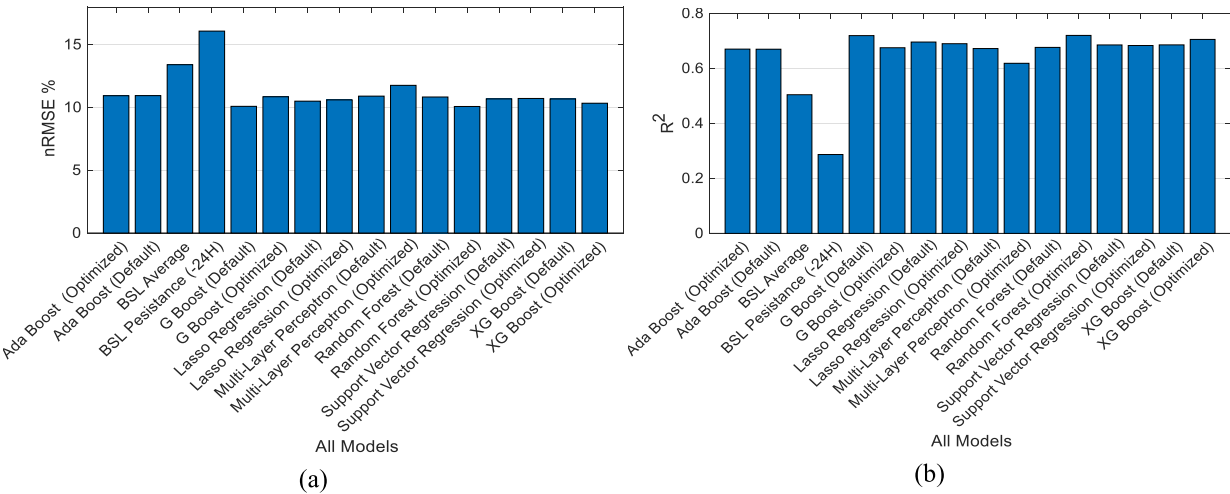


FIGURE 18. Combined performance evaluation of ML and baseline methods on testing data. (a) Normalized RMSE as percentage. (b) Coefficient of determination.

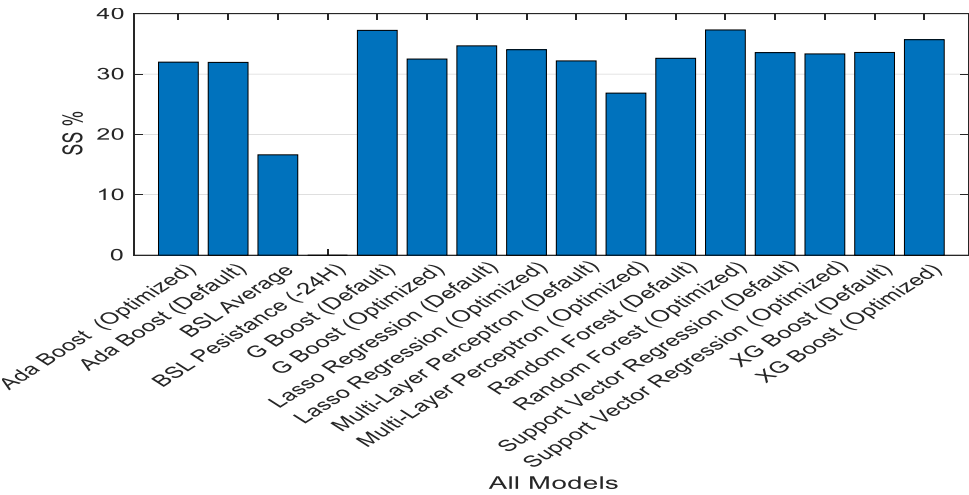


FIGURE 19. Combined performance evaluation of ml and baseline methods on testing data; skill score as percentage.

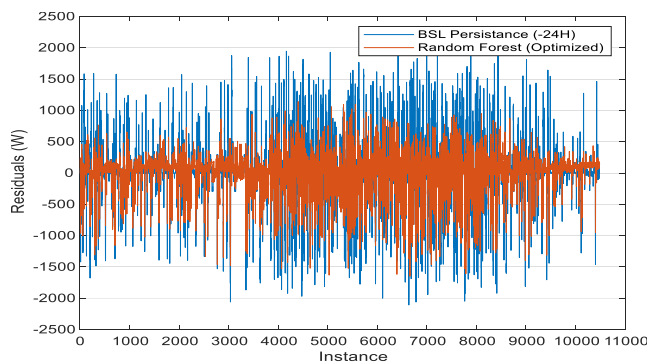
C. ML MODEL WITH GRID SEARCH HYPERPARAMETER OPTIMIZATION

After training and testing the ML models using default Scikit Learn library model Hyperparameters, new ML models were

retrained using grid search hyperparameter optimization. The results of the optimized models can be seen in TABLE 12 and Fig. 17. TABLE 12 is formatted such that an improvement in model forecasting performance is highlighted in green

TABLE 13. Combined methods evaluation.

Position	Model	MAE	MSE	RMSE	nRMSE (%)	R ²	SS (%)
1	Random Forest (Optimized)	143.81	63558.40	252.11	10.07	0.720	37.33
2	G Boost (Default)	132.76	63705.26	252.40	10.08	0.719	37.26
3	XG Boost (Optimized)	136.05	66875.34	258.60	10.33	0.705	35.71
4	Lasso Regression (Default)	152.49	69012.69	262.70	10.50	0.696	34.69
5	Lasso Regression (Optimized)	157.67	70375.20	265.28	10.60	0.690	34.05
6	XG Boost (Default)	143.43	71349.36	267.11	10.67	0.686	33.60
7	Support Vector Regression (Default)	161.47	71411.07	267.23	10.68	0.685	33.57
8	Support Vector Regression (Optimized)	163.90	71868.67	268.08	10.71	0.683	33.36
9	Random Forest (Default)	156.46	73448.30	271.01	10.83	0.676	32.63
10	G Boost (Optimized)	155.42	73714.20	271.50	10.85	0.675	32.51
11	Multi-Layer Perceptron (Default)	149.80	74386.84	272.74	10.90	0.672	32.20
12	Ada Boost (Optimized)	154.02	74843.30	273.58	10.93	0.670	31.99
13	Ada Boost (Default)	151.82	74933.31	273.74	10.94	0.670	31.95
14	Multi-Layer Perceptron (Optimized)	204.11	86565.51	294.22	11.75	0.619	26.86
15	BSL_Average (Climatology)	160.87	112458.63	335.35	13.40	0.504	16.64
16	BSL_Persistence (24H)	179.90	161819.59	402.27	16.07	0.287	0.00

**FIGURE 20.** Residuals of persistence and random forest forecast methods.

while a decline is highlighted in red. From TABLE 12, it is shown that the performance of other ML models improved after grid search optimization, such as for Random Forest, XG Boost and Ada Boost. However, the performance of the other ML models declined, such as that of G Boost and the rest. The decline in the performance of the other ML models may be caused by overfitting the ML models to the training data, hence the drop in performance when the models are tested on new data. When sorted according to SS, the Random Forest model performed the best, while the multi-layer perceptron NN was the least performing model.

D. COMBINED MODEL FORECASTS PERFORMANCE

TABLE 13, Fig. 18, and Fig. 19 show the results of all the tested forecasting models. From TABLE 13, when the performance of the forecasting models is ranked based on SS results, it shows that the grid search optimized Random Forest model is the best model with a SS of 37.33% and nRMSE of 10.07%. Apart from the baseline models, the worst-performing forecasting model is the optimized Multi-Layer Perceptron NN.

Fig. 20 and Fig. 21 provide a graphical representation of the comparisons of the best-performing model against the baseline model, where Fig. 20 shows a broad perspective of the residuals of the models across the test data and Fig. 21 shows a localized perspective of the model's performance through box plots of the model's absolute relative error for each hour of the day. It is evident from Fig. 20 that the optimized Random Forest model performs better than the baseline model consistently across most of the test data. This is shown by the residuals of the Random Forest being less than those of the persistence model in most of the forecasts made by the two models. When comparing the boxplots in Fig. 21, though the Random Forest model has more outliers than the persistence model, both the error bars and outliers for the Random Forest model are less than those for the persistence model except for the hours 1 to 7 and 18 to 24. This indicates that the Random Forest model is performing poorly in nighttime instances.

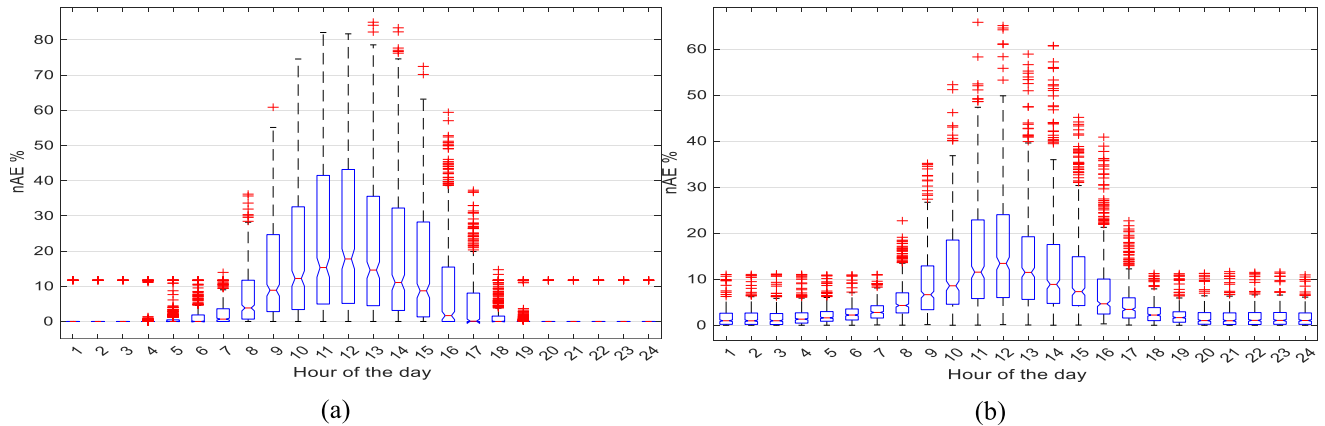


FIGURE 21. Boxplot for absolute relative error for each hour of the day. (a) Results for persistence baseline forecast method. (b) Results for random forest ML forecast methods.

VIII. CONCLUSION AND FUTURE WORKS

A. CONCLUSION

This paper presented a comprehensive and comparative review of existing papers on Machine Learning (ML) based approaches used in photovoltaic (PV) power forecasting. The work provided an overview of factors affecting solar PV power forecasting. These factors are weather conditions and forecasting horizons. From the weather conditions aspect, the effectiveness of solar PV systems is dependent on many factors including the amount of solar irradiance and ambient temperature which are subject to weather conditions. Accurate weather forecasting can be used to improve the ability to predict the power output of PV systems, and the meteorological variables that have the greatest impact on PV power can vary depending on the location of the PV plant. From the aspect of forecasting horizon, solar PV power forecasting can be categorized based on the forecasting time horizon, where the typical forecasting time horizons include very short term, short term, medium term and long term as shown in TABLE 1.

The paper further presented work providing an overview of ML, a description of several ML algorithms used in solar PV power forecasting and a description of commonly used performance metrics used for solar PV power forecasting. The presented ML algorithms are K-NN, SVM, Decision trees, Ensemble models such as random forest and ANN. The commonly used ML performance evaluation metrics are provided in chapter 4. Furthermore, an overview of related works is provided in TABLE 4, which shows some works from 2010 – 2020 in solar PV power forecasting.

Finally, a comparison of ML models based on simulations of solar PV forecasting was performed, where several ML solar PV forecasting models were created and compared against each other, and 2 baseline methods based on assumptions of persistence. In the simulations, the ML prediction models are trained on historical PV power production data and forecasted weather variables. The ML models created in the work include Gradient Boost (G Boost), extreme gradient Boost (XG boost), Support Vector Regression (SVR),

Random Forest (RF), Lasso Regression (LR), Ada Boost and A Multi-Layer Perceptron neural network (MLP NN). The baseline methods used in the experiment are 24 hours Persistence method and the time of day daily average method.

Using the solar PV power forecasting performance metrics presented in TABLE 9, findings from the simulation work show that ML models performed better than the baseline methods. This is evident from the ML models having lower RMSE and higher SS results than the baseline models as shown in TABLE 13. Furthermore, for the ML models with default hyperparameters, GBoost performed best with a SS of 37.326% while the RF model with optimized hyperparameters performed the best with a SS of 37.33% also shown in TABLE 13.

B. FUTURE WORKS

In the future, there are several areas that could be explored further in the field of photovoltaic (PV) forecasting. The accuracy of PV forecasting is essential for ensuring the reliable and efficient operation of the electric grid or off-grid PV systems. Numerical weather prediction (NWP) models are one of the key factors in the accuracy of PV forecasting. To improve PV forecasting, future work should focus on enhancing NWP models by incorporating additional data sources and utilizing advanced machine learning algorithms. Additionally, developing more sophisticated physical models of atmospheric processes could also improve the accuracy of PV forecasting.

As the use of PV systems for electricity generation continues to increase, there is an increasing need for accurate and reliable PV forecasting. This will require the development of advanced data preprocessing techniques that can handle large and complex datasets. In addition, AI models such as deep learning algorithms could be explored for use in tasks such as anomaly detection and outlier removal. Probabilistic forecasting techniques could also be applied to data preprocessing in PV forecasting.

Deep learning algorithms have demonstrated potential for use in PV forecasting but have not yet been widely adopted. Future work could focus on exploring different deep-learning architectures and algorithms to improve the accuracy and reliability of PV forecasting models. As the use of renewable energy grows, probabilistic forecasting will become increasingly important for power system operations. Future work should focus on developing probabilistic forecasting models that can accurately predict the uncertainty of PV power generation and incorporate this uncertainty into power system decision-making and risk management of solar PV power forecasting systems.

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